

Delivery Delay Analysis and Predictive Modeling Across Global Markets

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05/12/2025

Introduction

Global supply chains are facing continuous change, uncertainty, and reconfiguration due to wars, geopolitical tensions, and tariff policies. As a result, companies increasingly prioritize supply chain resilience to avoid loss of competitiveness, rising operational costs, and disruptions. Delivery reliability is a crucial indicator of logistics service quality and customer satisfaction. High delivery delay rates not only increase operational costs but also weaken a firm's competitive advantage. Therefore, digital tools—including machine learning, artificial intelligence (AI), and scenario modeling—have become essential strategies for providing real-time visibility and enhancing forecasting capabilities. Digitization in supply chains encompasses digital products, digital services, and the digital management of logistics processes within firms adapting to rapid technological change (Büyükoçkan and Göçer, 2018). To fully benefit from digital supply chains, it is necessary to understand how digitalization supports companies in improving operational performance. This study focuses on how machine learning can enhance delivery performance and reshape collaboration across supply chains.

Thus, this study addresses one main research question:

How can companies improve their delivery rates using machine learning?

To answer this question, the research integrates descriptive analytics and machine learning to provide both visibility and predictive capability for supply chain decision-making. The remainder of this study is organized as follows: Section 2 reviews the relevant literature, Section 3 explains the methodology and data collection process, Sections 4 and 5 present empirical findings, and the final section provides discussion and conclusions.

Literature Review

Transportation disruption is a critical risk in supply chains (Wilson, 2006), leading to late deliveries, operational shutdowns, lost sales, and reputational damage (Guiffrida and Jaber, 2008). Common delay drivers include shipment lead time and scheduling accuracy, carrier performance and transportation mode, operational bottlenecks within fulfillment centers, order-level anomalies such as fraud investigations or special handling, and market-level differences that may reflect infrastructure constraints. Based on these observations, the first hypothesis is formulated as follows:

H1: What is the relationship between the top five most important features and delivery performance?

To address delivery challenges, digital technologies such as machine learning and AI have emerged as powerful tools. Predictive analytics is transforming transportation from reactive to proactive decision-making, achieving 30–45% greater accuracy (Shavaki & Ghahnavieh, 2022). These technologies enhance firms’ ability to detect risks, anticipate disruptions, and increase resilience (Cui et al., 2023). In addition, given the globally distributed nature of supply chains, understanding delay rate variation across markets is critical. Thus, the second hypothesis is proposed:

H2: What is the relationship between delivery rates in the five markets with the highest delay rates?

However, limited research compares delay patterns across multiple global markets using large-scale transactional data while simultaneously integrating predictive modeling. This study aims to fill this gap.

Methodology

This study combines descriptive analytics and machine learning to analyze delivery delays and develop a real-time predictive model across global markets.

The dataset contains approximately 22,000 records, including customer, order, production, and transportation details, along with the labeled target variable *Late_delivery_risk*. A quantitative research design was adopted, consisting of three main phases.

First, descriptive analytics were conducted to compare delivery performance across different markets and identify whether delays stemmed from regional differences or systemic operational issues. Second, predictive modeling was performed using three supervised learning algorithms: Logistic Regression (L2 regularization), XGBoost, and Random Forest. Third, feature importance analysis was conducted using Random Forest to identify the top factors influencing delivery delays.

For data preprocessing, mode imputation was applied to categorical variables and mean imputation to numerical variables. Leakage variables—including *Late_delivery_risk*,

delivery status, scheduled and actual shipping days, and shipping dates—were removed. One-Hot Encoding and StandardScaler were applied to categorical and numerical variables, respectively.

Logistic Regression (L2 Regularization) was used to mitigate multicollinearity and enhance generalization while preserving interpretability. XGBoost, a gradient boosting model, was configured with 250 estimators, a learning rate of 0.1, a maximum depth of 5, and a subsample rate of 0.8. Random Forest was configured with 300 trees using Gini impurity as the splitting criterion.

An 80/20 train–test split was applied, and model performance was evaluated using Accuracy, F1 Score, and AUC. AUC served as the primary metric due to its robustness and ability to evaluate discriminative performance in class-imbalanced datasets.

Results

This chapter presents the empirical results of the study, including (1) descriptive statistics of delivery performance across global markets, (2) predictive model performance, and (3) feature importance analysis.

A total of 20,000 valid orders were analyzed. The overall late delivery rate was 55.2%, indicating that more than half of shipments failed to arrive on or before the scheduled delivery date.

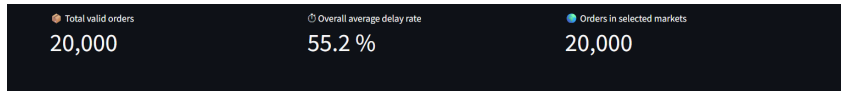


Figure 1: Overview

Table 1 summarizes the delay rate by market. Pacific Asia shows the highest delay rate at 56.35%, followed closely by Europe (55.52%) and LATAM (55.42%). Africa has the lowest rate at 52.23%, yet still reflects significant inefficiencies.

Table 1: Delay Rate by Market

Market	Orders	Delay Rate (%)
Africa	1,279	52.23%
USCA	2,925	53.95%
LATAM	5,749	55.42%
Europe	5,580	55.52%
Pacific Asia	4,467	56.35%

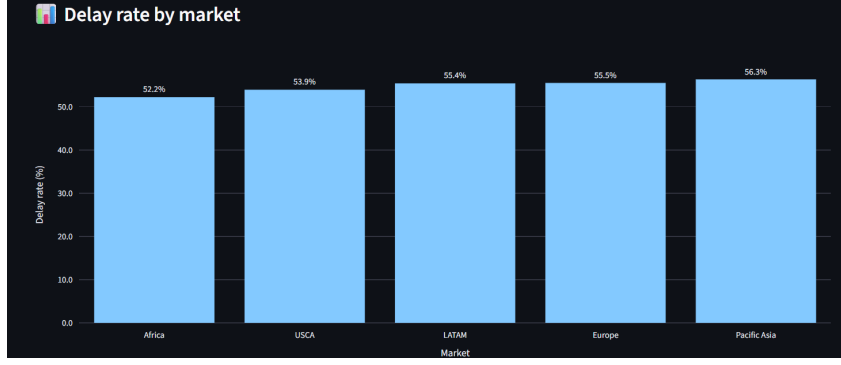


Figure 2: Delay Rate by Region

These results demonstrate that delivery delays are pervasive across the global supply chain network, highlighting the need for predictive modeling.

Next, predictive model performance. Three supervised machine learning models—Logistic Regression, Random Forest, and XGBoost—were trained using the 80/20 split. Table 2 reports the evaluation metrics. Logistic Regression achieved the highest AUC of 0.820, indicating strong discriminative performance. Despite being a linear model, it outperformed the nonlinear Random Forest and XGBoost models, suggesting that the predictors influence delivery delay risk in a largely linear and monotonic manner. Random Forest followed with an AUC of 0.802, while XGBoost achieved 0.781.

Table 2: Model Performance on 80/20

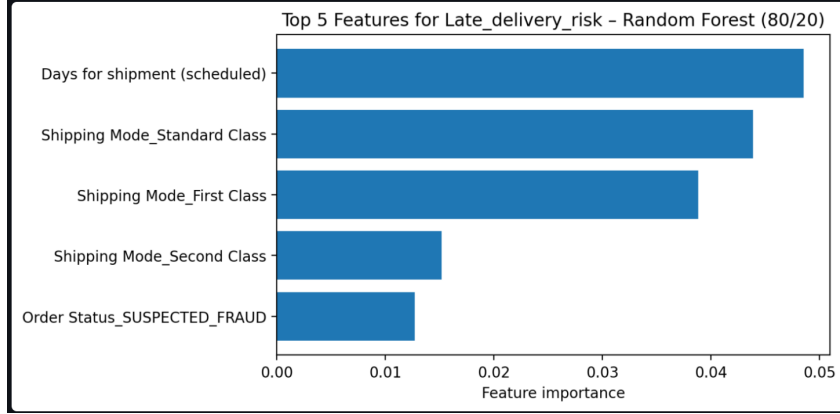
Model	Accuracy	F1 Score	AUC
Logistic Regression (L2)	0.738	0.754	0.820
XGBoost	0.712	0.699	0.781
Random Forest	0.726	0.713	0.802

Overall, Logistic Regression is the best-performing model and provides strong interpretability, making it suitable for managerial decision-making.

Feature importance analysis using Random Forest identified the top five predictors: scheduled shipment days, shipping mode (Standard, First, and Second Class), and order status classified as SUSPECTED_FRAUD. Longer scheduled shipment durations significantly increase the probability of late delivery, indicating inefficiencies in logistics planning. The influence of shipping mode highlights the importance of transportation service-level alignment. Fraud-suspected orders are more likely to be delayed due to additional verification or compliance reviews. Interestingly, market-level variables were not among the top predictors, suggesting that delays are driven by internal process factors rather than geographical conditions.

Table 3: Top 5 Important Features for Predicting Late Delivery Risk

Rank	Feature	Importance
1	Days for shipment (scheduled)	0.0485
2	Shipping Mode — Standard Class	0.0439
3	Shipping Mode — First Class	0.0388
4	Shipping Mode — Second Class	0.0152
5	Order Status — SUSPECTED_FRAUD	0.0127

**Figure 3:** Top 5 Important Features (Random Forest)

Discussion and Conclusion

This study analyzed global delivery delay patterns and developed machine learning models to predict late delivery risk. The findings offer several theoretical and managerial implications.

Regarding market-level delay patterns, all five regions—Africa, USCA, LATAM, Europe, and Pacific Asia—show delay rates exceeding 50%. Contrary to the assumption that delivery performance is primarily influenced by regional infrastructure or geographic challenges, the relatively small variation across regions indicates that systemic operational inefficiencies are the dominant drivers of delays.

Regarding model performance, Logistic Regression achieved the highest AUC and outperformed more complex models. This result suggests that interaction effects among variables are limited and that delivery delays follow a broadly linear pattern.

Feature importance analysis reveals three major insights. First, planning-related factors matter greatly. The strong influence of scheduled shipment days highlights inefficiencies in forecasting, network planning, or buffering strategies. Second, transportation mode selection is a critical determinant of delivery outcomes. Third, internal administrative processes—particularly fraud screening—introduce non-logistics delays, underscoring the role of internal governance mechanisms.

Managerially, this study suggests several actions: (1) improve shipment schedul-

ing accuracy by refining forecasting models and resource allocation; (2) optimise shipping mode selection based on customer priority, product characteristics, and required lead times; (3) strengthen carrier performance monitoring and diversify logistics partners where necessary; and (4) integrate ML-based risk detection systems into order management workflows to proactively flag high-risk orders.

In conclusion, this study provides a comprehensive assessment of global delivery delays and develops predictive models to estimate late delivery risk. The results show that delivery delays are widespread and likely stem from systemic inefficiencies. Logistic Regression demonstrates strong predictive power, while Random Forest feature importance highlights planning, transportation mode, and fraud-related processes as key delay drivers. These findings underscore the importance of improving logistics planning, transportation alignment, and proactive risk management through digital tools such as machine learning.

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